(9) Binomial Distribution

(1. Core Concepts:)

- **Binomial Trial (Bernoulli Trial) / Single Coin Flip:**

→ An experiment with two possible outcome:

- "success" (1): Probability = p.
- "failure" (0): Probability = $1-p$

→ A single binomial trial is like flipping a coin once

↳ Outcome 1 (success): "Heads" (arbitrarily chosen as our event of interest).

↳ Probability = p (e.g., $p=0.5$ for a fair coin).

↳ Outcome 0 (failure): "Tails"

↳ Probability = $1-p$.

Key Points:

- Only two possible outcomes per trial (binary).
- Probability p is constant for each trial.

- **Binomial Distribution / Multiple Coin Flips**

→ Describes the no. of successes (x) in 'n' independent trials.

→ The binomial distribution of getting exactly 'x' heads in 'n' flips?

Example:

- Experiment: Flip a fair coin 10 times ($n=10$, $p=0.5$).
- Question: What's the probability of getting exactly 3 heads ($x=3$)?

formula

$$P(X=x) = \binom{n}{x} p^x (1-p)^{n-x}$$

- $\binom{n}{x}$: Number of ways to choose 'x' successes in n trials (e.g. "10 choose 3").

Where $\binom{n}{x} = \frac{n!}{x!(n-x)!}$

Calculation

$$P(x=3) = \binom{10}{3}(0.6)^3(0.5)^7 = 120 \times 0.0009765625$$

$$= 0.117$$

→ There's an 11.7 chance of exactly 3 heads in 7 flip.

Key Takeaways

1. Single Trial = Coin flip (success / failure).
2. Binomial Distribution = Counts success in 'n' trials.
3. For Coin flips
   - ↳ $p = 0.5$: Symmetric distribution.
   - ↳ $p \neq 0.5$: Skewed (e.g $p = 0.1$ for rare events).

Real-world Examples

The Scenario

Imagine you run an online ad campaign.

- Each ad view is a trial (like a coin flip).
- Success: User clicks the ad (probability = $p = 0.02$).
- Failure: User ignores the ad (probability = $1 - p = 0.98$).

Binomial Probability in Action

"What is the probability of getting exactly 5 clicks in 100 views?

Formula:

$$P(x=5) = \binom{100}{5}(0.02)^5(0.98)^{95}$$

$$= 75,287,520 \times 0.000000000386 \approx 0.29 \ (29\%)$$

∴ There's a 29% chance of exactly 5 clicks in 100 views.

Cumulative Probability

"what is the probability of getting ≤ 5 clicks in 100 views?"

Soln

Sum probabilities for X = 0, 1, 2, 3, 4, 5:

$$P(X \leq 5) = \sum_{k=0}^{5} \binom{100}{k} (0.02)^k (0.98)^{100-k}$$

python code:

```
from scipy.stats import binom.

n = 100  # Ad views.
p = 0.02 # click probability.
cum_prob = binom.cdf (5, n, p).
print ("Probability of ≤ 5 clicks:
    {cum_prob:.2f}") # output: 0.92.
```

2. Key Properties.

Properties	formula	Explanation
probability Mass function (PMF)	$P(X=x) = \binom{n}{x} p^x (1-p)^{n-x}$	Probability of exactly x! success.
Cumulative Distribution Function (CDF)	$P(x \leq n) = \sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k}$	probability of x or fewer successes.
Mean (Expected Value)	$\mu = np$	Average number of successes.
variance	$\sigma^2 = np(1-p)$	spread of the distribution.

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3. When to use the Binomial Distribution

(1) Fixed number of trials (n).
(2) Independent trials (e.g. coin flips, survey responses)
(3) Constant success probability (p) for each trial.

Examples:
↳ click-through rate (CTR) prediction.
↳ loan default probability.
↳ Quality control ('defective items in a batch).

4. Understanding the Normal Approximation to Binomial Distribution.

1. Core idea

→ The binomial distribution (discrete) can be approximated by the normal distribution (continuous) under certain conditions. This simplifies calculation for large sample sizes.

2. When Does it Work?

Two conditions must be met.

(a) Moderate Probability:
↳ p should not be close to 0 or 1 (e.g. $0.1 \leq p \leq 0.9$)
↳ Why? Extreme 'p' values skew the distribution, making it non-symmetric.

(b) Sufficiently large Sample Size (n)
↳ Rule of thumb: $np \geq 10$ and $n(1-p) \geq 10$.
↳ Example: For $p = 0.5$, n should be at least 20 ($20 \times 0.5 = 10$).

③ Parameters of the Approximating Normal Distribution.

For a binomial distribution Binomial (n, p), the matching normal distribution has:

- Mean (μ) = np
  (e.g. 100 trials with $p = 0.3 \rightarrow \mu = 100 \times 0.3 = 30$).

- Standard Deviation (σ) = $\sqrt{np(1-p)}$
  (e.g. $\sigma = \sqrt{(100 \times 0.3 \times 0.7)} = \sqrt{21} \approx 4.58$.

④ Why These Condition?

- $np \geq 10$ and $n(1-p) \geq 10$: Ensure the distribution is symmetric and not skewed.
  ↳ if $p = 0.01$ (very small), you'd need $n \geq 1000$ to satisfy $np \geq 10$.

- Normal approximation fails if:
  ↳ p is extreme (e.g. $p = 0.001$).
  ↳ n is too small (e.g. $n = 5$, $p = 0.5 \rightarrow np = 2.5 < 10$

Exercise: Problem statement

no. of trials (n) = 200
probability of success (p) = 0.2.

Question: Approximate the probability of getting between 35 and 45 successes (inclusive) using the normal distribution.

Step 1: Verify Condition for Normal Approximation.
→ check if the binomial distribution meets the criteria for normal ~~distribution~~ approximation.

① Moderate p:
$p = 0.2$ (not extreme, since $0.1 \leq 0.2 \leq 0.9$) ✓

② Large enough n:
→ $np = 200 \times 0.2 = 40 \geq 10$ ✓
→ $n(1-p) = 200 \times 0.8 = 160 \geq 10$ ✓

Conclusion: Normal approximation is appropriate.

Step 2: Calculate Normal Distribution Parameters.

→ Convert the binomial distribution to a normal distribution

$\mu = 40$

$\sigma = \sqrt{[np(1-p)} = \sqrt{[200 \times 0.2 \times 0.8]} = \sqrt{32} = 5.66.$

So, the approximating normal distribution is:

$N(\mu = 40, \sigma = 5.66)$.

Step 3: Apply Continuity Correction.

→ Since the binomial distribute is discrete and the normal distribution is continuous, we adjust the boundaries by ± 0.05 for better accuracy. (± 0.05 for histogram bars)

- Lower bound: $35 \rightarrow 34.5$.
- Upper bound: $45 \rightarrow 45.5$.

Now, we calculate:

$P(34.5 \leq x \leq 45.5)$ in $N(40, 5.66)$.

Step 4: Calculate Z-scores.

→ Convert the bounds to standard normal (Z) scores:

$$Z = \frac{X - \mu}{\sigma}$$

- Lower bound (34.5):

$$Z = \frac{34.5 - 40}{5.66} = -0.97$$

- ~~Lower~~ Upper bound (45.5):

$$Z = \frac{45.5 - 40}{5.66} \approx 0.97$$

Step 5: Find Probabilities Using Z-Table.

Look up the probabilities for these Z-scores.

1. $P(Z \leq -0.97) = 0.1660$ (from Z-table).
2. $P(Z \leq 0.97) = 0.8340$.

Now, subtracting to get the probability between the bounds.

$$P(-0.97 \leq Z \leq 0.97) = 0.8340 - 0.1660 = 0.6680.$$

Interpretation: There approximately a 66.8% chance of getting between 35 and 45 successes.

5 (Key takeaways)

1. Use Binomial For:
   - Binary outcomes (success/failures).
   - Fixed n and constant p.

(2) Normal Approximation: valid for large n and moderate p.

(3) Mean/variance: $\mu = np$, $\sigma^2 = np(1-p)$.

6) Skewness

- The binomial distribution can be:
  - symmetric if $p = 0.5$
  - Right skewed if $p < 0.5$.
  - left-skewed if $p > 0.5$.

What does "Approximation" Mean?

→ An approximation is a representation of something that is not exact but is sufficiently close to be useful. For example, using 3.14 as an approximation for π (pi) which is approximately 3.14159, simplifies calculating while maintaining reasonable accuracy.

Poisson Approximation to the Binomial Distribution.

1) When to Use:
- n is large
- p is small
- The product $\lambda = n \times p$ is moderate.

2) Approximation:
Under these conditions, the binomial distribution $B(n, p)$ can be approximated by the Poisson distribution $P(\lambda)$, where $\lambda = n \times p$. This is particularly useful when calculating probabilities for rare events over a large number of trials.

3) Example
A factory produces $n = 10,000$ items with a defect rate of $p = 0.001$. What's the probability of finding exactly 12 defective items?

Step: Check Conditions
- $n = 10,000$ (large)
- $p = 0.001$ (small)
- $\lambda = n \times p = 10,000 \cdot 0.001 = 10$ (moderate)

☑ Poisson approximation applies.

Step 2 → Apply Poisson formula.

We want $P(x=12)$

$$P(x=12) = \frac{e^{-10} \cdot 10^{12}}{12!}$$

Calculations

1. $e^{-10} \approx 4.54 \times 10^{-6}$
2. $10^{12} = 1,000,000,000,000$
3. $12! = 479,001,600$

$$P(X=12) = \frac{(4.54 \times 10^{-6}) \times (1 \times 10^{12})}{4.79 \times 10^{8}} \approx 0.0948$$

Compare to Binomial (Exact) calculations.

$$P(X=12) = \binom{10,000}{12} (0.001)^{12} (0.999)^{9888}$$

$$\approx 0.0946$$

→ The Poisson approximation is very close and much easier to

4) Why Does This Work?

→ The Binomial distribution becomes computationally heavy for large n (e.g., $\binom{1000}{12}$ is enormous).

→ The Poisson distribution "ignores" n and focuses only on $\lambda = np$, making it efficient for many events.

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5.) Key Takeaways

(1) Use Poisson when:
↳ n is large, p is small, and $\lambda = np$ is moderate
↳ Example: Defects, call center arrivals, mutations in DNA.

(2) Do not use Poisson when
↳ p is large (e.g. $p > 0.1$).
↳ $\lambda = np$ is very large (use Normal approximation instead).

(3) Advantage: Simplifies calculations for rare events.

~~6) Rule of thumb for Better A:~~

Rule of thumb for Better Approximation

	Scenario	Best Approximation	Example
(1)	$n \geq 50$, $p \leq 0.1$, $\lambda \leq 10$	Poisson ($P(X=k) \approx e^{-\lambda}\lambda^k/k$	Defects in 10,000 items ($p = 0.001$, $\lambda = 10$).
(2)	$\lambda > 10$	Normal ($\mu = np$, $\sigma = \sqrt{np(1-p)}$)	500 successes in 10,000 trials ($p = 0.05$, $\lambda = 500$)
(3)	$n < 50$ or $p > 0.1$ (p not close to 0 or 1)	Exact Binomial ($\binom{n}{k} p^k (1-p)^{n-k}$)	2 heads in 5 coin flips ($p = 0.5$).

Rule of

(1) Poisson Approximation to the Binomial.

When to Use:

- Large number of trials: $n \geq 100$
- Small probability of success: $p \leq 0.1$
- Moderate expected number of successes:

$$\lambda = n \times p \leq 10.$$

Why Use it:

The Poisson distribution simplifies calculations when dealing with rare events over many trials. It's particularly useful when the exact binomial computation is complex due to large 'n' and small p.

Example

- Scenario: A factory produces 10,000 items with a defect rate of 0.001.
- Parameter: $n = 10,000$, $n = 0.001$ or $\lambda = 10$.
- Application: Use the Poisson distribution to approximate the probability of a certain number of defects.

(2) Normal Approximation to the Binomial

When to Use:

- Both np and $n(1-p)$ are greater than or equal to 10. This ensures the binomial distribution is approximately symmetric and bell-shaped.

Why Use it:

The normal approximation is beneficial when dealing with large sample sizes and probabilities not too

close to 0 or 1. It leverages the central limit Theorem to simplify calculations.

Example

- Scenario: Conducting 10,000 trials with a success probability of 0.05.
- Parameters: $n = 10,000$, $p = 0.05$, so $\mu = 500$, $\sigma = \sqrt{n \times p \times (1-p)} = \sqrt{10,000 \times 0.05 \times 0.95} = 21.79$.

(2) Exact Binomial Calculations.

When to use:

- small number of trials: $n < 50$
- Probability of success not extremely small or large: p not close to 0 or 1.

Why Use it:
When sample sizes are small, and probabilities are moderate, the exact binomial calculations provides precise probabilities without the need for approximation.

Example:

- Scenario: Flipping a fair coin 5 times.
- Parameters: $n = 5$, $p = 0.5$
- Applications: Calculate the probability of getting exactly 3 heads using the binomial formula.

$$P(X = 3) = \binom{5}{3} \times (0.5)^3 \times (0.5)^2 = 0.312$$

Python Code: Probability Mass Function (PMF)

```
import numpy as np.
import matplotlib.pyplot as plt.
from scipy.stats import binom.

# Parameters.
n = 10.
p = 0.5.

# Possible values of X
k_value = np.arange(0, n+1).

# PMF value
pmf_value = binom.pmf(k_value, n, p).

# Visualization
plt.stem(k_values, pmf_values)
plt.xlabel('Number of successes (k)')
plt.ylabel('Probability')
plt.title('Binomial PMF (n=10, p=0.5)').
plt.grid(alpha=0.3).
plt.show().
```

(1)

(A) Break-down

1) k_value = np.arrange(0, n+1).

what does this line do?

Imagine you're flipping a coin 10 times (n=10). You want to know:

- Probability of getting 0 heads
- Probability of getting 1 heads
- ----
- Probability of getting 10 heads.

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Python Code: Probability Mass Function (PMF).

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import numpy as np.
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# Parameters.
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(1)

(2) Break-down

1) k_value = np.arrange (0, n+1).

what does this line do?

Imagine you're flipping a coin 10 times (n=10). You want to know:

- Probability of getting 0 heads
- Probability of getting 1 heads
- ----
- Probability of getting 10 heads.

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k_values = np.arange(0, n+1) creates this list of all possible outcomes.

[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

why do we need this?

1.) for calculation
→ The computer needs to know which probabilities to calculate. binom.pmf(k_values, n, p) will compute the probability for each number in this list.

2) For Plotting → stem()
When we plot, we need:
- x-axis points: [0, 1, 2, ----, 10] (number of heads)
- y-axis points: The probabilities for each number of heads.

Concrete Example:
for coin flips (n=10, p=0.5).

→ k_values becomes [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10].
→ pmf_values will be:
- Probability of 0 heads: 0.001
- Probability of 1 heads: 0.010
- ...
- Probability of 10 heads: 0.001

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Mathematical PMF vs binom.pmf().

Q. Why $P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$ mathematical formula gives single and 'pmf-values = binom.pmf (k-values, n, p)' gives Array of multiple values?

⇒ 1. Mathematical Formula (Single Value).

$$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

↳ calculates probability for one specific k-value.

↳ Example for $k=5$

$$(X=5) = \binom{10}{5} (0.5)^5 (0.5)^5 = 0.240$$

⇒ 2. binom.pmf (k-values, n, p) (Array output).

↳ calculate probabilities for all k-values, in k-value simultaneously.

↳ Equivalent to running the PMF formula for each k in [0, 1, 2, ... 10].

↳ Returns an array where each entry corresponds to one k-value.

⇒ 3. Practical implications

- ↳ Single probability: Use binom.pmf (5, n, p) → return just 0.246093
- ↳ All probabilities: Use binom.pmf ([0, 1, ... 10], n, p) → returns the full array.

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Python Code: Cumulative Distribution function

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import binom

# Parameters
n = 10
p = 0.5

# Generate CDF values
k_values = np.arange(0, n+1)
cdf = binom.cdf(k_values, n, p)

print(cdf)
```

Approximating the Binomial CDF

Scenario	Best Approximation	Example
$n \geq 100$, $p \leq 0.1$, $\lambda \leq 10$	poisson Approximation	Defects in 10,000 with $p = 0.001$
$np \geq 10$, $n(1-p) \geq 10$	Normal Approximation	60 success in 100 trials with $p = 0.6$.
otherwise	Exact binomial calculation	small n or p not close to 0 or 1.

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1. Poisson Approximation.

When to use:

- Large number of trials: $n \geq 100$
- Small probability of success: $p \leq 0.1$.
- Moderate expected number of success: $\lambda = np \leq 10$.

Approximation:

The binomial CDF $P(X \leq k)$ can be approximated by the Poisson CDF:

$$P(X \leq k) \approx \sum_{i=0}^{k} \frac{e^{-\lambda} \lambda^i}{i!}$$

Example:

- Scenario: A factory produces 10,000 items with a defect rate of 0.001.
- Parameter: $n = 10,000$, $p = 0.001$, so $\lambda = 10$.
- Application: To find the probability of at most 5 defects, compute $P(X \leq 5)$ using the Poisson CDF with $\lambda = 10$.

Note: This approximation is particularly useful when dealing with rare events over many trials.

2. Normal Approximation:

When to use:

- Both np and $n(1-p)$ are greater than or equal to 10. This ensures the binomial distribution is approximately symmetric and bell-shaped.

Approximation:

The binomial CDF $P(X \leq k)$ can be approximated by the normal CDF with a continuity correction,

$$P(X \leq k) \approx \Phi\left(\frac{k + 0.5 - \mu}{\sigma}\right)$$

Where:

- $\mu = np$.
- $\sigma = \sqrt{np(1-p)}$.
- Φ is the standard normal CDF.

Example:

- Scenario: Conducting 100 trials with a success probability of 0.5.
- Parameters: $n = 100$, $p = 0.5$, so $\mu = 50$, $\sigma = 5$.
- Application: To find the probability of at most 60 successes, compute.

Note $$P(X \leq 60) \approx \Phi\left(\frac{60 + 0.5 - 50}{5}\right) = \Phi(2.1).$$

Note: The continuity correction (adding 0.5) adjust for the fact that the normal distribution is continuous, while the binomial is discrete.

Final Note:

Why Use Approximations instead of the Exact Binomial?

1. Computational Efficiency.
2. Analytical Simplicity.
3. Applicability in specific Scenarios.

1) Computational Efficiency.

→ Calculating exact binomial probabilities involves factorials computations, which can become computationally intensive as the number of trials (n) increases.

Approximations simplify these calculations, making them more tractable for large-scale problems.

2. Analytical Simplicity.

The binomial distribution is discrete, which can complicate analytical processes like integration or differentiation. The normal distribution, being continuous, allows for the application of calculus based methods, facilitating easier analysis and derivation of statistical properties.

3) Applicability in Specific Scenarios.
   - Poisson Approximation
   - Normal Approximation.

Binomial Differentiate: PMF vs CDF

Aspect	PMF	CDF.
Purpose	Probability of exact value ($P(X=k)$)	Probability up to a value ($P(X \leq k)$).
Computation	Single value calculation	sum of multiple PMF values up to k.
Graph Shape	Discrete bars at each k	step function increasing with k.
Use cases.	finding probability of a specific outcome	Determining probability within a range of outcomes.

Python code

Approximating the Binomial distribution using both the Poisson and Normal distribution in Python, calculating both PMF and CDF.

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```
Import numpy as np
from scipy.stats import binom, poisson, norm.
# Parameters-
  n = 100
  p = 0.05.
  k_values = np.arange (0, n+1) # Possible no. of successes.
# Binomial PMF and CDF.
binomial_pmf = binom.pmf (k_values, n, p).
binomial_cdf = binom.cdf (k_values, n, p).

  # Poisson Parameter
  lambda_poisson = n * p.
  # Poisson PMF and CDF
poisson_pmf = poisson.pmf (k_values, mu = lambda_poisson)
poisson_cdf = Poisson.cdf (k_values, mu = lambda_poisson)

  # Normal Parameters
  mu = n * p.
  sigma = np.sqrt (n * p * (1-p)).
  # Normal PMF approximation using the PDF.
normal_pmf = norm.pdf (k_values, loc = mu, scale = sigma)
  # Normal CDF approximation with continuity correction.
normal_cdf = norm.cdf (k_values + 0.5, loc = mu,
                       scale = sigma)
```

Note: In normal_cdf, k_values + 0.5 is known as the continuity correction. This adjustment is applied when using a continuous distribution, like the normal distribution, to approximate a discrete distribution, such as the binomial distribution